

2025 VCE Biology external assessment report

Specific information

This report provides sample answers or an indication of what answers may have included. Unless otherwise stated, these are not intended to be exemplary or complete responses.

The statistics in this report may be subject to rounding resulting in a total more or less than 100 per cent.

Section A

The table below indicates the percentage of students who chose each option. Grey shading indicates the correct answer.

Question	Correct answer	% A	% B	% C	% D	Comments
1	D	7	2	3	87	The independent variable is the variable that the experimenter changes.
2	C	5	3	88	4	Haemoglobin is a protein, so its activity is affected by temperature. The graph shows that activity decreases above 30°C as the protein is denatured, which disrupts secondary, tertiary and (in the case of haemoglobin) quaternary structures.
3	D	5	6	9	79	In both of the protein secretory pathways shown in the diagram, proteins are packaged into vesicles before being secreted at the plasma membrane. However, the pathways differ: Protein X moves directly from the rough endoplasmic reticulum to the plasma membrane, while Protein Y moves from the rough endoplasmic reticulum to the Golgi apparatus, which is responsible for modifying proteins, before reaching the plasma membrane.
4	B and C	2	38	53	7	The question states that 'The DNA triplet GCA codes for an amino acid': - Option B (tRNA anticodon GCA) was accepted as correct if the DNA triplet GCA is on the template (non-coding) strand, matching the mRNA codon CGU and corresponding tRNA anticodon GCA - Option C (tRNA anticodon CGU) was accepted as correct if the DNA triplet GCA is on the non-template (coding) strand, as the complementary template (coding) strand DNA triplet would be CGT, matching the mRNA codon GCA and corresponding tRNA anticodon CGU.
5	D	14	13	7	67	Option D is correct as RNA polymerase reaching a termination sequence on the DNA template is the last step of transcription. Option A is incorrect as the removal of introns is an event in mRNA processing (post-transcription modification). Options B and C are incorrect as these outline events in translation rather than transcription.

6	C	7	7	79	7	Option C is correct as section S could be any of the four sections of DNA and thus it could be part of a prokaryotic gene. Section R describes an operon, section Q belongs to a eukaryotic gene as it requires the removal of introns to form mature mRNA, and section T describes a regulatory gene.
7	C	6	3	90	2	Single guide RNA binds to the Cas9 protein and directs the Cas9 protein to where to cut the target DNA.
8	B	11	80	1	7	The gel electrophoresis shows only one larger fragment of DNA in condition 1, while conditions 2 and 3 each show two smaller DNA fragments, indicating that they have been cut at specific sites by CRISPR-Cas9, making option B the correct response.
9	D	3	7	9	81	Condition 1 contained only the Cas9 protein and target DNA, without the presence of either single guide RNA or its separate components (CRISPR RNA and trans-activating CRISPR RNA), and it was unable to cut the target DNA. Condition 2 contained both crRNA and tracrRNA and was able to cut DNA, as was the case with condition 3 where single guide RNA was formed by combining crRNA and tracrRNA. Therefore option D proposed a suitable hypothesis. Note that students were only required to recall the function of single guide RNA (sgRNA), not its two components (crRNA and tracrRNA).
10	B	13	75	3	8	The infographic shows the production of bioethanol, which is produced by anaerobic fermentation/respiration in yeast cells. Anaerobic fermentation in yeast cells involves glucose (in this case biomass such as waste starch) being broken down in the absence of oxygen to produce ethanol, carbon dioxide and two ATP molecules.
11	C	5	27	48	19	Students were required to carry out DNA profiling by interpreting a gel electrophoresis. Any DNA bands that are present in the lamb but not in the mother must have come from the father. By referring to the DNA bands of the four possible fathers, it can be determined that father A cannot be the lamb's father as the lamb's fourth largest band did not come from the mother and is not present in father A. Fathers B and D are also not the lamb's father as the lamb's shortest band (bottom band) did not come from the mother and is not present in fathers B or D. Only father C contains all three bands that are present in the lamb but not present in the mother.
12	B	3	87	7	2	Once an electric current is applied, DNA fragments travel through the gel from the negative to the positive end because DNA fragments are negatively charged.
13	D	12	11	5	72	Students were required to recall the steps involved in the polymerase chain reaction (PCR). Option A is incorrect as it is <i>Taq</i> /DNA polymerase that binds to the DNA template strand during extension. Option B is incorrect as DNA cools down in the annealing stage (after being heated in the denaturation stage). Option C is incorrect as DNA primers attach in the annealing stage. Option D is correct as the last stage of PCR is to repeat the cycle many times.
14	D	10	2	2	85	The reduction of leaf overlap will allow plants to absorb more light. Light is an input of the light-dependent stage, which occurs in the grana/thylakoid membranes, making option D correct. Option A is incorrect as the reduction of leaf overlap will

						not significantly increase absorption of carbon dioxide. Options B and C are incorrect as glucose and oxygen are outputs rather than inputs of photosynthesis.
15	B	21	54	15	10	Students were required to recall that the electron transport chain yields 26 or 28 ATP molecules per molecule of glucose, making option A incorrect. Students should also recall that there is variation between theoretical and actual yields of ATP (as outlined in the Frequently asked questions), making option B correct. Option C is incorrect as while root cells may not be able to carry out photosynthesis, they do have glucose from stored glucose/starch in the plant. Option D is incorrect as while C ₄ plants and C ₃ plants have some differences in their pathways for photosynthesis, they have the same cellular respiration pathway so they would produce the same amount of ATP per molecule of glucose.
16	C	6	9	66	18	Students needed to recall that yeast cells can undergo anaerobic fermentation/respiration by breaking down glucose in the absence of oxygen to produce ethanol, carbon dioxide and two ATP molecules. In some bacterial biochemical pathways, this ethanol is then further broken down into acetic acid to produce vinegar, making option C correct. Option D is incorrect as bacteria are prokaryotic organisms so they do not contain cristae, which is part of mitochondria.
17	B	19	63	9	8	Students needed to recall the inputs and outputs of the various stages of cellular respiration and photosynthesis. Option A is incorrect as stage T lists ATP as an output (not an input) of the light-independent stage. Option B is correct as stage U and V correctly identify inputs and outputs for the light-dependent stage and glycolysis, respectively.
18	A	70	5	10	15	Stage X represents anaerobic fermentation/respiration in animal cells, making option A correct. Options B and C are incorrect as ethanol and carbon dioxide are outputs of anaerobic fermentation in yeast rather than animals. Option D is incorrect as anaerobic fermentation occurs in the cytosol, not the mitochondria.
19	B	12	36	42	10	Students needed to be familiar with the investigation methodologies outlined in the key science skills section of the study design. Option A is incorrect as fieldwork would not be an appropriate methodology and stage Z identifies inputs and outputs of the electron transport chain (not photosynthesis). Option B is correct as a literature review would be an appropriate methodology to use to collate and analyse secondary data about the amount of NADPH produced in stage S; stage S also includes correct inputs and outputs for the light-dependent stage. Option C is incorrect as while a controlled experiment would be an appropriate methodology, glucose is an output (not an input) in photosynthesis. Option D is incorrect as while a computer model would be a suitable methodology, carbon dioxide is an input (not an output) of the light-independent stage within the stroma.
20	A	67	19	7	7	An increase in either light intensity or carbon dioxide concentration will result in the same trend, where the rate of photosynthesis will increase for a period of time and then plateau due to limiting factors, making option A correct. An increase in temperature will result in a different trend, where the rate of photosynthesis will first increase until optimum temperature is reached, then the rate of photosynthesis will

						decrease sharply as enzymes involved are denatured, making options B, C and D all incorrect.
21	D	43	9	15	34	ATP synthase is the enzyme. ATP is the product. ATP is formed from the combination of substrates ADP and P_i (inorganic phosphate). A competitive inhibitor binds to the active site of the enzyme, where the substrates bind, therefore preventing the substrates from binding. Option D is the correct option because the competitive inhibitor would not allow ADP to bind, resulting in a build-up of this within the cell. Option A is incorrect because the enzyme itself contains the active site, not the substrate or product.
22	A and B	26	51	19	4	Plants do not contain a second or third line of defence due to having no circulatory or lymphatic system. Plants do have a first line of defence (involving physical, chemical and microbiota barriers) to prevent the entry of and/or destruction of pathogens. The essential oils and phenolic compounds are included in this, therefore option B is correct. These are chemical antimicrobial agents that inhibit pathogen growth, therefore option A is also correct.
23	D	3	3	9	85	An anecdote is a short personal account of a personal, subjective event or observation. This best corresponds to Student 4 (option D) recalling how these essential oils and phenolic compounds were used in her childhood.
24	B	5	88	2	6	Option B is correct; cellular pathogens include bacteria and fungi, while viruses and prions are non-cellular pathogens and allergens including substances such as dog hair and pollen.
25	B	19	64	9	8	Swelling of the lymph nodes would be due to an accumulation of fluid or white blood cells in the area. This eliminates option A. Macrophages and dendritic cells present antigens on their MHC-II markers to T helper cells, making option C incorrect. In the lymph nodes, lymph fluid is screened and pathogens are removed, making option D incorrect. When a pathogen is detected, clonal selection and expansion of lymphocytes (specific T and B cells) occurs, therefore option B is correct.
26	D	3	1	1	94	Inflammatory responses lead to vasodilation (increased blood flow) and increased capillary permeability, which result in more white blood cells reaching the site of infection. This makes option D correct.
27	C	26	10	58	5	Students need to be familiar with the roles of the different molecules involved in the innate immune response. Histamines are released by mast cells and are involved in triggering the inflammatory response. Option A is incorrect as they do not act as enzymes. Option B is incorrect as interferons are produced by virally infected cells, but act to prevent the infection of other nearby uninfected cells. Option D is incorrect as lysozymes are produced by phagocytic cells to help in the destruction of pathogens. Option C is correct as complement proteins are usually inactive in the blood, and when activated they cause lysis of cells.
28	A	82	4	9	4	Plasma cells are a part of the adaptive humoral immune response. They secrete specific antibodies that bind to the antigens on extracellular pathogens. Therefore, option A is correct.
29	A	27	69	4	1	Antibodies are produced by plasma cells as part of the adaptive humoral immune response. When an individual is first exposed to an antigen, no existing antibodies or memory cells exist, and this third line of defence takes time to come into effect to produce antibodies. As the antibody concentration only begins to increase on

						day 13, the individual could not have been exposed on that day and must have been exposed earlier. Therefore, day 2 (option A) is the only correct answer.
30	D	25	8	4	63	Option A is incorrect because the individual has not been exposed to the antigen and cytotoxic T cells do not produce antibodies. Option B is incorrect, as plasma cells are formed from the differentiation of B cells, not T cells. After the first exposure to the antigen, memory cells are formed and upon subsequent antigenic exposure, these memory cells are reactivated leading to a faster and larger production of antibodies. This increase begins to be visible from day 35, therefore the activation could have taken place just prior to this (day 33) resulting in the faster antibody production, making option D correct.
31	A	90	6	4	0	A large proportion of the population vaccinated results in herd immunity. This reduces the number of available hosts and subsequently the transmission of the pathogen between individuals, supporting option A.
32	C	5	7	84	4	Option C is correct because if individuals in a population travel, they may encounter new pathogens in other locations and bring these back to their own population, resulting in the emergence of this pathogen within their own population.
33	D	6	13	2	79	Bacterial mutations that result in antibiotic resistance do not often affect antigen structure, so new vaccines would not need to be developed, eliminating option B. Antigenic drift is the result of minor alterations in viral antigens that may still allow existing antibodies and memory cells to recognise the virus, so a new vaccine would not be required, eliminating option A. Antigenic shift results in the combination of genetic material from multiple viruses, leading to new antigens which would not be recognised by existing antibodies and memory cells. As such, a new vaccine would need to be developed in this situation, making option D correct.
34	B	2	86	2	10	Typically, phylogenetic trees are used to compare the relatedness of species or populations based on molecular homology. When interpreting phylogenetic trees, the most recent node you can trace back to when comparing two species is when they share the most recent common ancestor. Therefore, Labradors are equally related to German Shepherds and Basenjis, so option D is not correct. Option B is correct because species that are more closely related will have fewer differences in their molecular homology (DNA or amino acids).
35	B	16	75	7	2	The presence of a tail and fur covering the body would not help distinguish this lemur as a primate from other mammals who also possess a tail. However, not all mammals have an opposable thumb, so this classifies the lemur as a primate, making option B correct.
36	C	7	9	79	6	Fossils that contain characteristics of two different types of organisms, such as reptiles and mammals in this example, are called transitional fossils. These fossils indicate that the two share a common ancestor at some point in the evolutionary journey, so option C is the correct answer. These fossils are not always complete, so cannot show all characteristics of the organisms, making option B incorrect.
37	B	1	80	16	2	The image shows the time periods (age) of some layers. The younger the age, the higher the layer is, and vice versa. Therefore, fossil X, which is only present in layer S, must be between the ages of 145 and 165 mya. Thus, option B is correct. Students should note that index fossils are useful when abundant and present in many locations, and exist only in one time period, thus making option C incorrect.

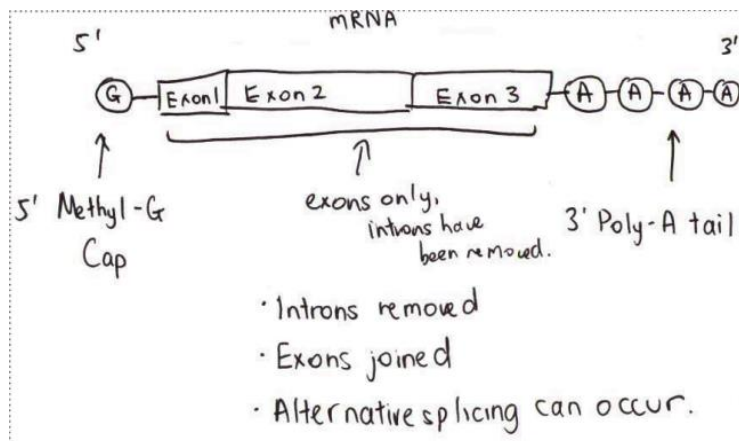
38	A	69	5	13	13	Students must remember to use all the information provided in the question; in this case it was stated the pig-nosed turtle was found only in the Northern Territory and was hypothesised to have recently arrived in Australia. This specific hypothesis, which related to time of arrival, can only be disputed by evidence that would indicate it was not a recent arrival, such as finding five-million-year-old fossilised remains, as referred to in option A. Evidence of vestigial or homologous structures and molecular homology would not dispute this specific hypothesis of when the pig-nosed turtle arrived in Australia.
39	D	15	2	3	80	The Year 12 student experiment was investigating the effects of different amounts of water (independent variable, what is being changed) on the growth of lettuce plants (dependent variable, what is being measured). The experiment would be valid if there was only one independent and one dependent variable, with all other variables controlled. However, from the table of results, temperature has not been controlled for each lettuce plant, making option D correct. As temperature is an additional independent variable, not a dependent variable, option A is incorrect.
40	C	9	6	81	3	Qualitative data is observational and can be subjective, compared to quantitative data which is numerical, objective and can be measured. All of concentration, number and mass are measurable units of data, making these quantitative. Therefore, the shape of the lettuce leaves in option C is the only qualitative data present.

Section B

Question 1a.

Mark	0	1	2	3	Average
%	25	5	14	56	2.01

The diagram should be replicated with an indication that either introns had been removed OR exons spliced together, as shown in the sample answer below:



Two additional modifications that could have been drawn and labelled included:

- methyl/modified guanine cap
- poly-A tail
- alternative splicing (or a specific example such as intron retention or exon juggling).

The thin lines connecting exons 1 and 2, and exons 2 and 3 in the original diagram were introns. Students were required to recognise that these non-protein coding regions are removed as a result of mRNA processing (post-transcription modification).

Students were required to recognise that the modified guanine cap (methyl cap) and the poly-A tail should be added to opposite ends of the final mRNA strand. However, they were not required to recall that these are added to the 5' and 3' end, respectively.

Students are encouraged to write in blue or black pen, even when drawing diagrams, to ensure answers scan clearly.

Question 1b.

Mark	0	1	2	Average
%	46	30	24	0.78

In Step 4, DNA ligase joins the phosphodiester bonds between the human insulin gene A or B and their respective plasmid.

In Step 8, amino acid chains A and B are combined to produce a functional insulin protein or quaternary structure.

Students commonly misunderstood that each of the insulin genes are inserted separately into their respective plasmid and are not inserted together into the same plasmid.

Question 2a.

Mark	0	1	2	Average
%	44	33	23	0.79

Enzymes have an active site complementary to specific substrate(s), which bind to that active site. Enzymes lower the activation energy required to convert substrates into products.

As the question required students to explain **how** enzymes catalyse biochemical reactions, marks were not awarded for only stating that enzymes catalyse or speed up reactions.

Question 2b.

Mark	0	1	2	3	4	5	Average
%	21	12	17	16	19	14	2.43

Students were required to make reference to both timeframes referred to in the question, recognising that between 0 and 20 minutes, *trp* was present in the cell and therefore enzyme activity was low as represented by the graph. After 60 minutes, *trp* was absent and therefore enzyme activity was steadily high as represented by the graph.

Responses could have included:

- Between 0 and 20 minutes when *trp* is present:
 - repression and/or attenuation are occurring
 - the rate of transcription is low because the *trp* repressor periodically detaches from the operator
- After 60 minutes when *trp* is absent and enzyme activity is constant:
 - repression and/or attenuation are not occurring
 - attenuation prevents more enzymes from being synthesised
 - enzymes already produced for *trp* synthesis are saturated or other limiting factors were present.
- Repression and attenuation work together to regulate the production of enzymes:
 - when *trp* levels are high
 - to prevent transcription and/or translation
 - to prevent *trp* being synthesised
 - through attenuation occurring when repression fails.

In many cases, students provided detailed knowledge of the mechanisms of repression and attenuation in both scenarios where *trp* was present and absent. However, accurate knowledge of these regulation mechanisms is required for the responses to be awarded full marks. Common mistakes included stating the repressor binds to the promoter or RNA polymerase was not able to translate genes, or confusing the terminator and anti-terminator hairpin loop.

Students are also reminded to read the axis labels carefully. In this question, students often confused the y-axis as representing *trp* concentration as opposed to enzyme activity.

Answering in dot points is acceptable and can help students to ensure they are addressing all parts of a question. Students should use the number of marks as an indication of the level of detail required in their response.

Question 2c.

Mark	0	1	2	Average
%	38	43	19	0.8

Examples of limitations for the model simulation could have included that the simulation:

- is overly simplified
- does not account for what occurs in real time or in all prokaryotes
- assumes ideal conditions or does not consider other factors that may have changed, such as temperature or pH
- represents a general trend that excludes outliers
- does not represent the results observed in the two sets of experimental data or enzyme activity above 1.0 (which is seen in the experimental data)
- only accounted for when *trp* concentration was present or absent, rather than when this concentration would vary.

This question was commonly misinterpreted by students who provided limitations of the two sets of experimental data (shown by triangles and squares) as opposed to the limitations of the model simulation.

Example of a high-scoring student response:

- *The model was overly simplified and did not replicate real world random variations in trp levels in the cell.*

Question 2d.

Mark	0	1	2	Average
%	57	38	6	0.49

In this response, students were required to identify a consequence for the bacteria now containing a codon for *trp* (as opposed to an additional stop codon) and justify their response. Students needed to associate this different codon with an impact on protein formation, structure and/or function and a consequence for the bacteria as a result.

Examples of impacts on protein formation, structure and/or function could have included:

- polypeptide sequence is longer or contains additional *trp* amino acids
- polypeptide chains will fold differently
- different proteins/enzymes produced
- previously produced proteins may no longer be functional.

Examples of appropriate consequences, either positive or negative, for the bacteria could have included:

- additional amino acids were required to be synthesised or obtained from the environment
- attenuation (or mechanisms of this process) would now be required or tRNA exists carrying *trp*
- additional energy was required by the cell
- less chance of nonsense mutations occurring
- increased or decreased bacterial survival.

Example of a high-scoring student response:

- *Previous genes with the UGG codon at the end would now code for a longer primary structure of the protein as the UGG codon now codes for trp, rather than signalling termination of translation.*
- *Changes in primary structure may alter proteins in these bacteria.*
- *Lack of functional proteins disadvantageous for survival of bacteria.*

Question 3a.

Mark	0	1	2	3	Average
%	3	21	63	14	1.87

	Location of Rubisco activity		Time of day when Rubisco activity is greatest	
	Mesophyll cells	Bundle sheath cells	Daytime	Night-time
C₃	X		X	
C₄		X	X	
CAM	X		X	

Students were awarded 1 mark for every two boxes correct. No mark was awarded when students put a cross in both cell locations or both times of day for a specific plant type.

Question 3b.

Mark	0	1	2	Average
%	33	17	50	1.16

With increasing temperature, C₃ plants do not have any adaptations to reduce photorespiration or may close their stomata. Oxygen is more likely to bind with Rubisco as opposed to carbon dioxide.

In their response, students needed to indicate that it is specifically Rubisco which fixes oxygen and carbon dioxide (not carbon), rather than just the plant/cell in general. Statements about denaturation were not awarded marks. Students should also recognise that if denaturation occurred at higher temperatures, it would cause a permanent irreversible alteration of Rubisco's active site, meaning neither oxygen nor carbon dioxide would bind successfully.

Spelling is not penalised in written responses in Biology exams, unless the misspelled word is a different correct word (e.g. if students misspelled 'stoma' as 'stroma' in this question).

Question 3ci.

Mark	0	1	2	Average
%	12	32	57	1.45

Examples of how light could be controlled could have included:

- plants placed at an equal distance from the light/lamp OR light intensity/wavelength measured or kept the same
- timing how long plants were exposed to light
- removing sources of ambient or additional light.

Examples of how water could be controlled could have included:

- measuring a set volume of water or watering plants at the same time of day
- regulating humidity or soil moisture.

In this question, responses were required to describe **how** the students could control both light and water for the plants under investigation. Simple statements such as 'providing the same amount of light' or 'controlling the amount of water' did not address how this was performed, and simply restated information provided in the stem of the question. Responses should avoid use of the term 'amount' and instead use more specific terminology, such as 'volume', when discussing water.

Question 3cii.

Mark	0	1	2	Average
%	37	46	17	0.8

Reasons why measuring the amount of glucose would not be appropriate for the students' experiment could have included:

- difficult to measure glucose
- glucose is not a product (output) of photorespiration
- glucose may be consumed (input) for cellular respiration
- glucose can be converted into starch/cellulose in plants or used in the formation of fruits/seeds
- plants may need to be damaged to test for or obtain glucose.

This question highlights the importance of students knowing the inputs and outputs of photosynthesis and cellular respiration. Glucose is not an input to photosynthesis or photorespiration. If students referenced respiration using glucose, they must be aware that the term 'respiration' does not allow enough distinction to know if they were referring to photorespiration or cellular respiration in this question.

Question 4a.

Mark	0	1	2	Average
%	15	41	45	1.29

Crop	Transgenic, genetically modified or both	Justification
rice	Genetically modified	The (<i>OsH XK1</i>) gene in rice has been edited without inserting a gene from a different species
maize	Both	The maize genome was edited by inserting genes from a different species (<i>Bacillus thuringiensis</i>) into it and all transgenic organisms are genetically modified

Question 4b.

Mark	0	1	2	Average
%	22	31	47	1.25

Explanation of how editing of the *OsH XK1* gene in rice could result in an increased crop yield could have included:

- more light energy absorbed
- increased rate of photosynthesis
- greater production of glucose.

Some responses simply restated that chlorophyll levels in the leaves were increased, rather than connecting increased chlorophyll levels to the ability of the leaves to absorb more light energy or to carry out more photosynthesis. These responses were not awarded a mark. Other responses about increased growth rate were also not awarded a mark, as they needed to be connected to how this would occur (e.g. by producing more glucose).

Question 5a.

Mark	0	1	2	Average
%	24	32	44	1.19

Roles of neutrophils in an immune response to *Neisseria meningitidis* could have included:

- migration to the site of infection
- release of cytokines to attract other immune cells
- phagocytosis of bacterium
- release of toxins/defensins that are toxic to bacterium.

Question 5b.

Mark	0	1	2	3	4	Average
%	41	12	16	18	13	1.48

Contrasts between how the adaptive immune system response to an extracellular pathogen, like *N. meningitidis*, and an intracellular pathogen, like the influenza virus, could have included:

- the type of immunity involved in each response differs: humoral immunity is involved in a response to an extracellular pathogen, whereas cell-mediated immunity is involved in a response to an intracellular pathogen
- the type of cells involved in each response differs: plasma B cells respond to extracellular pathogens, while cytotoxic T cells respond to intracellular pathogens
- the molecules and mechanisms involved in targeting each type of pathogen are different: antibodies agglutinate/neutralise/opsonise extracellular pathogens, whilst cytotoxic granules (perforin/protease) induce death of cells infected by intracellular pathogens
- the type of memory cells involved in each response is different: B memory cells are produced in response to extracellular pathogens, whereas T memory cells are produced in response to intracellular pathogens.

Example of a high-scoring student response:

- Immune system utilises the humoral response to respond to extracellular pathogens whereas the cell-mediated response responds to intracellular pathogens.*
- Humoral response results in the production of plasma cells to produce antibodies whereas the cell-mediated response produces cytotoxic T cells.*
- Antibodies make it easier for the immune system to recognise and destroy pathogens via opsonisation whereas cytotoxic T cells induce apoptosis in virally infected cells.*
- Cell-mediated response produces T memory cells whereas humoral response produces B memory cells.*

It was evident that many students did not understand how the adaptive immune response differed towards extracellular versus intracellular pathogens. Some students confused extracellular pathogens with pathogens that had not yet entered the body and incorrectly made contrasts between first, second and third lines of defence.

Some students included similarities between the two responses. Students need to be familiar with the various command terms that appear in VCE Biology examinations; 'contrast' requires a student to outline differences, not similarities. Refer to the [VCAA Glossary of command terms](#) for more detail.

Where available, students are advised to write in the empty space underneath a question, such as the space on p. 31 of the exam booklet below the prescribed lines, rather than continuing their answer on the 'Extra space for responses' pages at the back of the exam paper. This extra space at the back of the exam booklet should only be used where there is insufficient space to finish a response directly under the question.

Question 6a.

Mark	0	1	2	Average
%	32	27	40	1.07

The correct answer was natural passive immunity:

- passive because babies receive antibodies OR do not produce their own antibodies and/or memory cells
- natural because these antibodies are provided from the mother (through the placenta/breastmilk) OR the baby obtained the antibodies through no further medical intervention.

Students must ensure they understand that active and passive are different strategies and contradict each other. Likewise for natural and artificial/induced immunity.

Question 6b.

Mark	0	1	2	Average
%	29	37	34	1.05

Reasons why a booster vaccine is advised if the mother became pregnant again could have included:

- antibodies and/or memory cells decline over time
- faster and/or larger antibody and/or memory cell production is required
- to ensure the baby receives sufficient antibodies
- to protect the baby/mother from RSV.

Responses often incorrectly discussed memory cells being passed down from mother to child, which is incorrect. Additionally, many students did not make reference to either the baby or the mother in their response, instead providing pre-planned, generic responses about booster vaccines.

Responses referring to antigenic drift were not awarded marks, as this may change the RSV antigens, rendering the booster vaccine ineffective. A new vaccine would be required.

Question 6c.

Mark	0	1	2	Average
%	17	18	65	1.47

The economic benefit from the vaccination of pregnant women is that the relevant stakeholder either saves money (e.g. hospital, government, society) or makes money (e.g. pharmaceutical companies).

Examples of explanations that could lead to economic benefit could have included:

- reduced hospitalisations or emergency care or paediatric treatment
- cheaper to vaccinate compared to treating preventable diseases
- lowered risk of RSV or other infections/illness
- reduced strain on government-funded healthcare programs or reduced burden on public health system
- increased herd immunity preventing community spread of RSV
- reduced maternal sick leave or less carer/partner absences from workplaces
- improved long-term child health or reduction in child health/education support later in life.

Generic responses such as 'children survived so they could contribute to the economy' were not accepted as a valid economic benefit. Many responses referred to saving costs by vaccinating two individuals with the one vaccine. Whilst this is true prior to birth, it was not awarded marks when the responses continued on to state that the child would then not require vaccination after being born, as this is biologically incorrect.

For this type of question, students are encouraged to apply their knowledge and skills of ethical understanding (see page 12 of the study design). Specifically, students should be able to apply the VCE Biology Units 1–4 key science skill regarding '... the influence of social, economic, legal and political factors relevant to the selected issue', including how these factors may affect decision-making and consequences of these decisions.

Question 6d.

Mark	0	1	2	Average
%	54	20	26	0.71

Explanations of how monoclonal antibodies work could have included:

- binding to (RSV) antigen (not binding to pathogen/protein)
- agglutination of virus
- neutralisation of virus or prevention of virus spread
- opsonisation or attraction of phagocytes to RSV
- activation of complement proteins
- activation of natural killer or cytotoxic T cells (which could occur through bispecific monoclonal antibodies, mAbs)
- delivery of drugs/treatment directly to pathogen-infected cells (conjugated mAbs)
- reduction of symptoms/number of RSV/number of infected cells.

This question was challenging for students as it required them to apply their knowledge of the use of mAbs to an immune response against a pathogen such as RSV, rather than demonstrate their understanding in the more familiar contexts of the use of mAbs in the treatment of cancer and autoimmune diseases. The question required students to recognise that mAbs can function like any other antibody. Therefore, two suitable actions of antibodies were required.

There were misconceptions by some students that antibodies kill pathogens or attach to antigens on pathogen-infected cells, whereas they actually attach to antigens of extracellular pathogens and trigger other responses to occur to eliminate the pathogen. Responses outlining how mAbs were produced were not awarded marks.

Example of a high-scoring student response:

- *Monoclonal antibodies have complementary variable regions to antigens on the RSV virus.*
- *Thus they can bind to viruses and mark them for phagocytosis, or clump them together to immobilise the pathogen, enhancing the speed and strength of the immune response especially for infants without fully developed adaptive immune systems, thus reducing the severity of symptoms.*

Question 6e.

Mark	0	1	2	Average
%	20	18	61	1.41

Aside from administering a vaccination (and the herd immunity that arises from this) or monoclonal antibody, ways to reduce the spread of RSV, with suitable justifications, could have included:

Identify one way to reduce spread in infants	Justification
- wear masks (nurses, infants, mothers, visitors, doctors, etc.) - mothers avoid large gatherings - quarantine/isolate infected infants	Blocks/reduces respiratory droplets. OR Prevents person-to-person transmission
- disinfect surfaces - wash hands with soap - improve hygiene practices - do not share equipment/rooms/beds	Stops virus being transferred from one surface to another. OR Prevents person-to-person transmission.
education (on how the virus is spread)	So doctors/nurses/mothers/visitors take precautions to prevent transmission. OR Encourages people to apply relevant strategy (e.g. from the 'identify' column, such as wearing masks and disinfecting surfaces).

Students needed to think carefully about their choice of strategy. For example, it would not be appropriate or ethical to isolate or quarantine the child from the mother. Some students identified an action and then justified this by simply stating 'reduces the spread' – such a response was simply restating the stem of the question instead of providing a justification and was not awarded a mark.

Students needed to use the context provided, remembering that information at the beginning of Question 6 still applies to part e.

Question 7a.

Mark	0	1	2	Average
%	31	38	31	0.99

The founder effect:

- occurs when a small group of individuals establishes a new population with reduced genetic variation compared to the original population.

The scientists are correct because:

- only a single colony of Asian honey bees were introduced to Australia
- only the genetic material of the queen and the sperm she had stored from her native environment were introduced to the new colony.

Many responses did not make reference to genetic change, as was required by the question. Some students wrote about a smaller/unrepresentative population being produced, without explaining how this occurred, or without linking it to a smaller gene pool or unrepresentative allele frequencies. Some responses incorrectly described the bottleneck effect instead of the founder effect.

Question 7b.

Mark	0	1	2	3	Average
%	30	27	34	9	1.22

The process responsible for the DNA differences is mutations.

Explanations of how this process contributed to the successful expansion of the colonies could have included:

- new alleles arose, increasing genetic diversity
- individual bees with alleles that resulted in traits which increased their chances of survival were selected for
- over time, this led to an increased proportion of individual bees with favourable alleles.

Example of a high-scoring student response:

- Mutations.*
- Mutations cause the formation of new alleles in the honeybee genome by altering the base sequence of DNA.*
- Some of these alleles are highly beneficial, providing a selective advantage to individuals that have them, causing those with the alleles to have an increased likelihood of survival and reproduction, passing on favourable alleles to offspring, over generations producing entire colonies with the advantageous alleles which aids in successful expansion.*

Many responses did not directly answer the 'identify the process responsible for these DNA differences' part of the question. If they did, natural selection or genetic drift were common incorrect answers. These would change the frequency of alleles rather than actually producing the DNA differences that led to the new alleles.

Question 7c.

Mark	0	1	2	3	Average
%	27	26	31	15	1.34

Explanations of potential genetic change in the native bee population could have included:

- the genetic diversity in the native bee population may decline as alleles may be lost
- the native bee population may reduce in numbers, reducing its gene pool
- individual native bees that cannot compete with the Asian honey bees are selected against and do not survive (or vice versa)
- these native bees do not reproduce, so less advantageous alleles are not passed on and decrease in frequency.

Example of a high-scoring student response:

- *Native bees may see a reduction in genetic diversity as only those bees able to survive and compete with Asian bees (selection pressure) for food and colony space will be able to reproduce and pass on these traits.*
- *Hence some alleles may see a decrease in frequency or be removed altogether as only advantageous traits increase in allele frequency with disadvantageous traits (worse food collection etc.) reduce in frequency or are erased from the gene pool altogether hence decreasing genetic diversity.*

Responses should avoid Lamarckian statements, such as suggesting the native bees adapted in order to survive competition from Asian honey bee species. Students should understand that genetic variants must be present for the species or population to survive threats from other species.

Students are encouraged to relate their answers to the context of the question, rather than simply regurgitating pre-planned responses.

Question 8a.

Mark	0	1	2	Average
%	6	40	55	1.49

Reasons why tadpoles were chosen for this experiment could have included:

- tadpoles are laid in large groups, making it easier to obtain a large sample size for the experiment
- tadpoles have transparent skin, meaning that light could easily pass through and be absorbed by the cyanobacteria
- tadpoles have gills and live in water, therefore they can absorb dissolved oxygen
- tadpoles are small animals, so they are easy/cheap to look after
- tadpoles have a fast growth/reproduction rate, therefore the time to collect results is short
- use of tadpoles received ethical approval as they are less complex animals.

Lower-scoring responses simply listed information from the article, without proposing a connection between this information and why tadpoles were chosen for this experiment.

Question 8bi.

Mark	0	1	2	Average
%	28	55	18	0.89

The molecules that photosynthetic cyanobacteria would require to produce oxygen are:

- water
- chlorophyll
- coenzymes/NADP/ADP + P_i.

Students need to be familiar with the inputs and outputs of the light-dependent and light-independent stages of photosynthesis. Some responses incorrectly identified carbon dioxide and/or glucose, which are inputs/outputs of the light-independent stage and not directly involved in the production of oxygen. Light was also not awarded a mark, as while it is required to produce oxygen, it is not a molecule.

Question 8bii.

Mark	0	1	2	Average
%	35	37	28	0.93

Role of each of the molecules named in part b.i. in the production of oxygen:

- water is split to form oxygen and load coenzymes/NADP/ADP + P_i
- chlorophyll captures light energy to facilitate splitting of water and loading of coenzymes/NADP/ADP + P_i
- coenzymes/NADP/ADP + P_i assist enzymes to function by transferring electrons/energy/hydrogen ions between reactions.

A mark was awarded for this question if the student incorrectly identified light as a molecule in part b.i. (with no mark awarded for part b.i.) but then provided a correct description of the role of light energy in the splitting of water to produce oxygen in part b.ii.

Question 8c.

Mark	0	1	2	Average
%	23	58	20	0.96

Explanations of why the tadpoles' brain activity stopped could have included:

- tadpoles could not carry out aerobic respiration as oxygen is required as an input
- tadpoles carried out anaerobic respiration, which produces less ATP than aerobic respiration
- cells were unable to produce sufficient ATP/energy.

Many students identified that the dark, hypoxic environment meant photosynthesis was not occurring and therefore oxygen was not being produced, however they were unable to connect this to aerobic/anaerobic respiration and how it would affect the tadpole's brain activity. Some responses stated that low oxygen led to denaturation of enzymes or death of cells, however this was incorrect as brain activity was able to be restarted.

Question 8d.

Mark	0	1	2	Average
%	38	30	31	0.93

Reasons for the delay in the restart of the tadpoles' brain activity could have included:

- it takes time for oxygen to be produced
- it takes time for photosynthesis to occur
- it takes time for oxygen to be used
- It takes time for aerobic/cellular respiration to occur
- it takes time for ATP to be produced
- oxygen/ATP is required for brain activity.

Example of a high-scoring student response:

- *After cyanobacteria were exposed to light, time was taken for the photolysis of water to occur and oxygen (O₂) levels to reach sufficient amounts for cellular respiration to occur (which is also time consuming and slow) in order to produce sufficient ATP for the brain to start functioning again.*

Students needed to refer to a biochemical pathway in their response, connecting the pathway to the production of oxygen or ATP. Low-scoring responses simply identified a biochemical pathway or outlined steps in photosynthesis or cellular respiration without connecting it to the tadpoles' brain activity.

Question 8e.

Mark	0	1	2	Average
%	27	23	50	1.22

Events that occur along the electron transport chain could have included:

- coenzymes/NADH/FADH₂ transfer/unload electrons/energy/hydrogen ions
- water is formed
- oxygen is used as an input
- 26 or 28 ATP molecules are produced.

As outlined in the study design, students need to be familiar with the inputs and outputs of the various stages of cellular respiration, including ATP yield (details of biochemical pathway mechanisms are not required).

Question 8f.

Mark	0	1	2	Average
%	8	23	69	1.6

Justifications using information from the article to support agreement with Student X could have included:

- beneficence relates to maximising benefits and reducing harm/risk
- benefit is being maximised as the research could save human lives in future by increasing oxygen levels in medical emergencies
- harm is being minimised as while tadpole brain activity was stopped, it was restarted again.

Example of a high-scoring student response:

- *Student X.*
- *Beneficence involves maximising positive outcomes (or ensuring if harm is inevitable, that it is outweighed).*
- *Positive outcomes such as the potential of increasing oxygen levels in humans during medical emergencies were maximised, saving lives, rendering it ethical in the long term despite minimal harm to tadpole.*

OR

Justifications using information from the article to support agreement with Student Y could have included:

- non-maleficence relates to avoiding harm OR ensuring harm is not disproportionate to benefits
- benefits are disproportionate as the potential positive outcome of improving treatment for humans during medical emergencies does not outweigh the harm being caused to the tadpoles
- harm is not being minimised as tadpoles are being harmed by having cyanobacteria injected into their hearts.

Example of a high-scoring student response:

I agree with student Y. Treating tadpoles as test subjects and intentionally placing them in hypoxic conditions may cause stress and pain. This violates non-maleficence (the minimisation of preventable harm) as the tadpoles are intentionally placed in distress from hypoxic conditions.

Some responses included justifications that did not match the student (X or Y) they were agreeing with. Students need to be able to demonstrate and apply ethical understanding using the ethical concepts outlined on pages 15 and 16 of the study design.

Question 9a.

Mark	0	1	2	Average
%	30	33	37	1.07

Examples of molecular homology comparisons could have included:

- nucleic acid (nucleotides, DNA, RNA) sequences
- amino acid sequences.

A description of how molecular homology can be interpreted could be:

- the more similar these sequences are between two populations or the less time for mutations to occur between them, the more closely related they are (or vice versa).

As this question was about different populations of woodpecker finches rather than different species, students should be careful about using standard responses that discuss how molecular homology can be used to compare species, as opposed to populations.

Question 9b.

Mark	0	1	2	3	Average
%	13	35	27	26	1.65

The type of speciation could have included:

- allopatric
- geographic isolation
- finches on San Cristobal may not be able to fly the distance to other islands.

Explanations of how speciation occurred could have included:

- no gene flow/interbreeding between populations
- different mutations accumulating in each population
- each population experiencing different selection pressures and certain individuals having a selective advantage.

Students did not need to use the term 'allopatric speciation' in their response; instead they could indicate their understanding of the type of speciation by referring to the population as geographically isolated. As students were required to explain the type of speciation that occurred, they needed to state more than just 'different selection pressures were acting'. Making reference to a survival/selective/reproductive advantage or individuals containing certain alleles/phenotypes/traits conferring an advantage was important.

This was not sympatric speciation as there was no evidence of different mating times of the finches or other temporal isolations. and the woodpecker finches on San Cristobal island were not living in the same location as the other woodpecker finch populations.

Question 10a.

Mark	0	1	2	Average
%	38	35	27	0.88

Evidence that would support Conclusion 1 needed to relate to a feature of the footprints that would demonstrate bipedalism. This could have included:

- non-opposable big toe
- arched foot
- more pronounced heel bone.

Evidence that would support Conclusion 2 needed to relate to a feature that differs between different bipedal footprints. This could have included:

- angle of the big toe compared to other toes is different
- different size of arch in foot
- different distribution of weight from heel to toe.

For Conclusion 2, students needed to identify a structural difference between the footprints rather than simply stating that the footprints would be different or that they would be different sizes, as this could indicate the footprints were from the same species but of different ages (child/adult size) rather than from different species.

Question 10b.

Mark	0	1	2	Average
%	16	30	54	1.38

Reasons for the difference in the interbreeding conclusions made, with a suitable justification using information from the table, could have included:

- there were different numbers of genomes studied, as Study 1 involved 7 genomes while Study 2 involved 334 genomes
- the individuals studied were from different time periods, as Study 1 involved individuals dated 45 000 years ago while Study 2 involved individuals dated from both 45 000 to 2200 years ago and from present day
- the two studies had different levels of accuracy, as Study 2 had a larger sample size than Study 1 and/or Study 2 involved both ancient and present-day individuals whereas Study 1 only involved ancient individuals
- researchers can interpret data differently as demonstrated by the different ranges of interbreeding found in the two conclusions, with Study 1 concluding that the range of interbreeding was from 49 000 years ago to 45 000 years ago while Study 2 concluded that the range of interbreeding was from 50 500 years ago to 43 500 years ago.

Question 10c.

Mark	0	1	2	Average
%	23	40	37	1.14

Explanations for the change in the percentage of Neanderthal DNA found in ancient human individuals compared to present-day individuals could have included:

- there has been a decrease in the percentage of Neanderthal DNA in the genomes of present-day humans
- more time has passed since *Homo sapiens* and *Homo neanderthalensis* shared a common ancestor OR more time for mutations to accumulate in the DNA of present-day individuals compared to ancient humans
- in present-day individuals there has been more time for interbreeding between *H. sapiens* OR between *H. sapiens* and other species of the genus *Homo*
- *H. sapiens* and *H. neanderthalensis* can no longer interbreed as *H. neanderthalensis* is extinct
- present-day individuals may not be descended from ancestors with high levels of Neanderthal DNA
- over time, different selection pressures have favoured different alleles, resulting in alleles that were prominent in ancient humans being lost.

When discussing interbreeding, students needed to provide specific details, such as which species were interbreeding, when they were interbreeding or to what extent they were likely to interbreed.

Question 10d.

Mark	0	1	2	Average
%	40	27	32	0.92

Benefits of using mtDNA to help determine the pathways hominin species may have taken could have included:

- mtDNA is passed down the maternal line or mtDNA is not subject to recombination
- mtDNA has a known/high mutation rate
- mtDNA's only source of variation is mutation
- mtDNA is more abundant or has a higher copy number than nuclear DNA.

Responses that described how mtDNA can determine migratory pathways rather than outlining the benefits of using mtDNA were not awarded marks.